

RESEARCH INTERESTS:

Stratospheric dynamics, satellite remote sensing, ozone and radiative forcing, stratosphere-troposphere exchange, atmospheric general circulation, and climate variability.

EDUCATION:

University of Washington, Seattle, WA September 2022 – Present
Atmospheric Science Major
| Master's degree in Atmospheric and Climate Science June 2025
| Ph.D. in Atmospheric Science Expected Graduation: June 2027

Washington College, Chestertown, MD August 2018 – May 2022
Physics and Mathematics Double Major
| Bachelor of Sciences (Phi Beta Kappa)

RESEARCH EXPERIENCE:

Graduate Research Assistant September 2022 – Present
Department of Atmospheric and Climate Science, University of Washington

Advisor: Dr. Qiang Fu

| Investigating the role of large-scale atmospheric circulation in controlling stratosphere–troposphere exchange of air mass and ozone using satellite observations, reanalyses, and chemistry–climate model simulations.

Oral Presentation(s): AMS conference (January 2023, January 2024), the International Conference on GPS Radio Occultation (ICGPSRO, October 2023), and the GNSS-Remote Sensing Colloquium (GNSS-RSC, June 2023).

Poster Presentation(s): QBOi-Snap-QUOCA Workshop (QSQ, March 2025)

CISESS Summer Intern June 2025-September 2025
Cooperative Institute for Satellite Earth System Studies (University of Maryland)

Mentors: Xi Shao, Feng Zhang, Changyong Cao

Project Title: Ensuring Stability and Consistency of AMSU-A Across Legacy NOAA Missions using AI/ML

| Developed a machine-learning pipeline using NOAA-15 AMSU-A brightness temperatures and reanalysis collocations to reconstruct atmospheric thermodynamic profiles, supporting cross-sensor continuity for microwave sounding data records and numerical weather prediction applications.

Award(s): Best CISESS Summer Intern Presentation Honorable Mention

Undergraduate Intern June 2021 – August 2021
William Lapenta Internship, National Oceanic and Atmospheric Association, Silver Spring, MD

| Studied temperature inversions associated with tropospheric ducting and validated COSMIC II satellite data against widely accepted weather models.

Oral Presentation(s): AMS conference (January 2022)

TEACHING and ADVISING:

Graduate Teaching Assistant September 2025 – December 2025
Department of Atmospheric and Climate Science, University of Washington September 2023 – December 2023

Courses: Introduction to Global Warming (ATM S 111), Climate and Climate Change (ATM S 211)

PUBLICATIONS and PAPERS:

1. **Hall, A.**; Dong, C.; Fu, Q.; Solomon, S., (2026) Interannual Variability in the Stratosphere-Troposphere Exchange of Air Mass and Ozone in Chemistry Climate Models. *in Preparations*
2. Mudhar, R.; **Hall, A.**; Huang, J.; Lee, R.W.; Palmeiro, F.M. QBOI - SNAP - QUOCA Workshop: Improved Simulations of the Stratosphere for Better Predictions of Weather, Climate and Extreme Events. *Weather* **2026**, *81*, 15–17, doi:10.1002/wea.7788.
3. **Hall, A.**; Fu, Q.; Dong, C. Revisiting the Stratosphere–Troposphere Exchange of Air Mass and Ozone Based on Reanalyses and Observations. *Atmosphere* **2025**, *16*, 1050, doi:10.3390/atmos16091050.
4. Wang, M.; Fu, Q.; **Hall, A.**; Sweeney, A. Stratosphere-Troposphere Exchanges of Air Mass and Ozone Concentrations From ERA5 and MERRA2: Annual-Mean Climatology, Seasonal Cycle, and Interannual Variability. *JGR Atmospheres* **2023**, *128*, e2023JD039270, doi:10.1029/2023JD039270.

LEADERSHIP and OUTREACH:

UW Women in Atmos Group	2023-Present
<i>President</i>	
Graduate Climate Conference	2024
<i>Organizer</i>	
UW Depart of Atmospheric and Climate Science Outreach	2022-2024
<i>Coordinator</i>	

TECHNICAL SKILLS:

Python, Machine Learning, Fortran, ncl